

Commerce XI Maths App

Theory (Pages 1-7)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 50

Question 1 - Let's Recall P1

When observations are arranged in ascending order, the middle value is called the

Options

- A. median
- B. mode
- C. mean
- D. range

Correct answer: median

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives median.

Question 2 - Let's Recall P1

The median divides an ordered data set into

Options

- A. two equal parts
- B. three equal parts
- C. four equal parts
- D. ten equal parts

Correct answer: two equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives two equal parts.

Question 3 - Let's Recall P1

For odd n , the position of the median is

Options

- A. $(n + 1)/2$
- B. $n/2$
- C. $(n - 1)/2$
- D. $2n$

Correct answer: $(n + 1)/2$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(n + 1)/2$.

Question 4 - Let's Recall P1

For even n , the median is the mean of observations at positions

Options

- A. $n/2$ and $(n + 2)/2$
- B. $(n + 1)/2$ only
- C. $n/4$ and $3n/4$
- D. 1 and n

Correct answer: $n/2$ and $(n + 2)/2$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $n/2$ and $(n + 2)/2$.

Question 5 - Theory P2

Values that divide ordered data into a required number of equal parts are called

Options

- A. partition values
- B. class marks
- C. frequencies
- D. deviations

Correct answer: partition values

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives partition values.

Question 6 - Theory P2

The median is a partition value because it divides the data into

Options

- A. two equal parts
- B. four equal parts
- C. ten equal parts
- D. one hundred equal parts

Correct answer: two equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives two equal parts.

Question 7 - Theory P3

The three principal types of partition values studied in this unit are

Options

- A. quartiles, deciles and percentiles
- B. mean, median and mode
- C. range, variance and deviation
- D. frequency, class and tally

Correct answer: quartiles, deciles and percentiles

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives quartiles, deciles and percentiles.

Question 8 - Theory P3

Quartiles divide ordered observations into

Options

- A. four equal parts
- B. two equal parts
- C. ten equal parts
- D. one hundred equal parts

Correct answer: four equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives four equal parts.

Question 9 - Theory P3

The number of quartiles is

Options

- A. 3
- B. 2
- C. 4
- D. 10

Correct answer: 3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 3.

Question 10 - Theory P3

The first or lower quartile is denoted by

Options

A. Q_1

B. Q_2

C. Q_3

D. D_1

Correct answer: Q_1

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_1 .

Question 11 - Theory P3

The second quartile is denoted by

Options

A. Q_2

B. Q_1

C. Q_3

D. P_2

Correct answer: Q_2

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_2 .

Question 12 - Theory P3

The third or upper quartile is denoted by

Options

A. Q_3

B. Q_1

C. Q_2

D. D_3

Correct answer: Q_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_3 .

Question 13 - Theory P3

The percentage of observations below quartile Q_1 is

Options

- A. 25%
- B. 50%
- C. 75%
- D. 10%

Correct answer: 25%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 25%.

Question 14 - Theory P3

The percentage of observations below quartile Q_2 is

Options

- A. 50%
- B. 25%
- C. 75%
- D. 20%

Correct answer: 50%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 50%.

Question 15 - Theory P3

The percentage of observations below quartile Q_3 is

Options

- A. 75%
- B. 25%
- C. 50%
- D. 90%

Correct answer: 75%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 75%.

Question 16 - Theory P3

Which identity is correct?

Options

- A. $Q_2 = \text{Median}$
- B. $Q_1 = \text{Median}$
- C. $Q_3 = \text{Median}$
- D. $Q_1 = Q_3$

Correct answer: $Q_2 = \text{Median}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_2 = \text{Median}$.

Question 17 - Theory P3

For raw data containing n observations, the position of Q_i is

Options

- A. $(i(n+1))/4$
- B. $(iN)/10$
- C. $(i(n-1))/4$
- D. $((n+i))/4$

Correct answer: $(i(n+1))/4$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i(n+1))/4$.

Question 18 - Theory P3

If a partition position is fractional, the textbook method uses

Options

- A. linear interpolation between adjacent observations
- B. rounding down only
- C. rounding up only
- D. the class midpoint

Correct answer: linear interpolation between adjacent observations

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives linear interpolation between adjacent observations.

Question 19 - Theory P3

Before calculating quartiles for raw data, observations must be arranged in

Options

- A. ascending order
- B. random order
- C. descending order only
- D. frequency order

Correct answer: ascending order

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ascending order.

Question 20 - Theory P3

For grouped data, the quartile class contains the

Options

- A. $(iN)/(4)$ th observation
- B. $(iN)/(10)$ th observation
- C. $(iN)/(100)$ th observation
- D. $(N + i)$ th observation

Correct answer: $(iN)/(4)$ th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(iN)/(4)$ th observation.

Question 21 - Theory P3

The formula of quartiles for grouped data is

$$Q_i = L + (h)/(f) \left((iN)/(4) - \text{c.f.} \right), i = 1, 2, 3$$

L denotes the

Options

- A. lower boundary of the quartile class
- B. upper boundary
- C. largest observation
- D. total frequency

Correct answer: lower boundary of the quartile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives lower boundary of the quartile class.

Question 22 - Theory P3

The formula of quartiles for grouped data is

$$Q_i = L + \frac{(h)}{(f)} \left(\frac{(i N)}{(4)} - \text{c.f.} \right), i = 1, 2, 3$$

h denotes the

Options

- A. class width of the quartile class
- B. frequency of the quartile class
- C. cumulative frequency
- D. class mark

Correct answer: class width of the quartile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives class width of the quartile class.

Question 23 - Theory P3

The formula of quartiles for grouped data is

$$Q_i = L + \frac{(h)}{(f)} \left(\frac{(i N)}{(4)} - \text{c.f.} \right), i = 1, 2, 3$$

f denotes the

Options

- A. frequency of the quartile class
- B. total frequency
- C. preceding cumulative frequency
- D. class width

Correct answer: frequency of the quartile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives frequency of the quartile class.

Question 24 - Theory P3

The formula of quartiles for grouped data is

$$Q_i = L + \frac{(h)}{(f)} \left(\frac{(i N)}{(4)} - \text{c.f.} \right), i = 1, 2, 3$$

c.f. denotes the

Options

- A. less than cumulative frequency of the class just preceding the quartile class
- B. frequency of the quartile class
- C. total frequency
- D. upper boundary

Correct answer: less than cumulative frequency of the class just preceding the quartile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than cumulative frequency of the class just preceding the quartile class.

Question 25 - Theory P3

The formula of quartiles for grouped data is

$$Q_i = L + \frac{(h)(f) \left(\frac{iN}{4} - \text{c.f.} \right)}{f}, i = 1, 2, 3$$

N denotes the

Options

- A. total frequency
- B. frequency of the quartile class
- C. class width
- D. lower boundary of the quartile class

Correct answer: total frequency

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives total frequency.

Question 26 - Theory P3

The central 50% of observations lies between

Options

- A. Q_1 and Q_3
- B. Q_1 and Q_2
- C. Q_2 and Q_3
- D. D_1 and D_9

Correct answer: Q_1 and Q_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_1 and Q_3 .

Question 27 - Solved Example P3

For $n = 19$, the position of Q_1 is

Options

- A. 5th observation
- B. 4.75th observation (interpolated)
- C. 19th observation
- D. 6th observation

Correct answer: 5th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 5th observation.

Question 28 - Solved Example P3

For $n = 19$, the position of Q_2 is

Options

- A. 10th observation
- B. 9.5th observation (interpolated)
- C. 5th observation
- D. 11th observation

Correct answer: 10th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 10th observation.

Question 29 - Solved Example P3

For $n = 19$, the position of Q_3 is

Options

- A. 15th observation
- B. 14.25th observation (interpolated)
- C. 5th observation
- D. 6.33th observation (interpolated)

Correct answer: 15th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 15th observation.

Question 30 - Solved Example P4

For $n = 12$, the position of Q_1 is

Options

- A. 3.25th observation (interpolated)
- B. 3rd observation
- C. 12th observation
- D. 4.25th observation (interpolated)

Correct answer: 3.25th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 3.25th observation (interpolated).

Question 31 - Solved Example P4

For $n = 12$, the position of Q_2 is

Options

- A. 6.5th observation (interpolated)
- B. 6th observation
- C. 3.25th observation (interpolated)
- D. 7.5th observation (interpolated)

Correct answer: 6.5th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 6.5th observation (interpolated).

Question 32 - Solved Example P4

For $n = 12$, the position of Q_3 is

Options

- A. 9.75th observation (interpolated)
- B. 9th observation
- C. 3.25th observation (interpolated)
- D. 4th observation

Correct answer: 9.75th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.75th observation (interpolated).

Question 33 - Theory P5

For $n = 100$, the position of Q_1 is

Options

- A. 25.25th observation (interpolated)
- B. 25th observation
- C. 100th observation
- D. 26.25th observation (interpolated)

Correct answer: 25.25th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 25.25th observation (interpolated).

Question 34 - Theory P5

For $n = 100$, the position of Q_2 is

Options

- A. 50.5th observation (interpolated)
- B. 50th observation
- C. 25.25th observation (interpolated)
- D. 51.5th observation (interpolated)

Correct answer: 50.5th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 50.5th observation (interpolated).

Question 35 - Theory P5

For $n = 100$, the position of Q_3 is

Options

- A. 75.75th observation (interpolated)
- B. 75th observation
- C. 25.25th observation (interpolated)
- D. 33.33th observation (interpolated)

Correct answer: 75.75th observation (interpolated)

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 75.75th observation (interpolated).

Question 36 - Solved Example P3

For 19 ordered observations, Q_1 is the value of the

Options

- A. 5th observation
- B. 4th observation
- C. 10th observation
- D. 15th observation

Correct answer: 5th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 5th observation.

Question 37 - Solved Example P3

For 19 ordered observations, Q_2 is the value of the

Options

- A. 10th observation
- B. 5th observation
- C. 9th observation
- D. 15th observation

Correct answer: 10th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 10th observation.

Question 38 - Solved Example P3

For 19 ordered observations, Q_3 is the value of the

Options

- A. 15th observation
- B. 5th observation
- C. 10th observation
- D. 14th observation

Correct answer: 15th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 15th observation.

Question 39 - Solved Example P4

For the ordered wages 170, 180, 190, 200, 210, 220, 280, 310, 320, 330, 380, 400, Q_1 equals

Options

- A. 192.5
- B. 190
- C. 195
- D. 200

Correct answer: 192.5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 192.5.

Question 40 - Solved Example P4

For the ordered wages 170, 180, 190, 200, 210, 220, 280, 310, 320, 330, 380, 400, Q_2 equals

Options

- A. 250
- B. 220
- C. 245
- D. 280

Correct answer: 250

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 250.

Question 41 - Solved Example P4

For the ordered wages 170, 180, 190, 200, 210, 220, 280, 310, 320, 330, 380, 400, Q_3 equals

Options

- A. 327.5
- B. 320
- C. 325
- D. 330

Correct answer: 327.5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 327.5.

Question 42 - Practice P1–7

The daily earnings (in ₹) of 12 workers are 420, 380, 510, 460, 390, 540, 480, 430, 570, 410, 490, 450.

Which method should be used to calculate the quartiles?

Options

- A. Raw-data observation-position method
- B. Grouped-data quartile formula
- C. Less-than cumulative-frequency method
- D. Graphical method only

Correct answer: Raw-data observation-position method

Explanation and steps

These are 12 individual observations, not class intervals or cumulative frequencies. Arrange the earnings in ascending order and use the raw-data observation positions $i(n + 1)/4$. Therefore, use the raw-data observation-position method.

Question 43 - Practice P1-7

Why can the discrete-data method be used for the following distribution?

Number of books (X)	1	2	3	4	5
No. of students (f)	3	7	9	6	5

Options

- A. The exact value corresponding to every observation is known
- B. The data are given in continuous class intervals
- C. The values of the observations are unknown
- D. The total frequency is less than 50

Correct answer: The exact value corresponding to every observation is known

Explanation and steps

Each frequency is attached to a known value of X (1, 2, 3, 4 or 5). Hence every observation has an exact value and the discrete-data method applies.

Question 44 - Practice P1-7

Calculate Q_2 for the following discrete frequency distribution.

X	5	7	9	11	13
f	2	4	5	6	3
c.f.	2	6	11	17	20

Correct answer: 9

Explanation and steps

$N = 20$, so the Q_2 position is $2(N + 1)/4 = 10.5$. The cumulative frequency first reaches this position at c.f. = 11, corresponding to $X = 9$. Therefore, $Q_2 = 9$.

Question 45 - Practice P1-7

Calculate Q_3 for the following discrete frequency distribution.

X	20	25	30	35	40	45
f	3	4	5	8	2	2
c.f.	3	7	12	20	22	24

Correct answer: 35

Explanation and steps

$N = 24$, so the Q_3 position is $3(N + 1)/4 = 18.75$. The cumulative frequency first reaches this position at c.f. = 20, corresponding to $X = 35$. Therefore, $Q_3 = 35$.

Question 46 - Practice P1-7

The following is a less-than cumulative frequency distribution.
Find the frequency of the 30–40 marks group.

Marks below	20	30	40	50	60
No. of students	8	23	47	76	100

Correct answer: 24

Explanation and steps

In a less-than cumulative table, a class frequency is found by subtraction. Frequency for 30-40 = number below 40 - number below 30 = $47 - 23 = 24$.

Question 47 - Practice P1-7

The following is a less-than cumulative frequency distribution.
Find the frequency of the 70–80 kg group.

Weight below (kg)	50	60	70	80	90
No. of persons	5	18	42	73	90

Options

- A. 31
- B. 17
- C. 24
- D. 3

Correct answer: 31

Explanation and steps

Frequency for 70-80 kg = number below 80 kg - number below 70 kg = $73 - 42 = 31$.

Question 48 - Practice P1-7

For the following grouped distribution, $N = 120$ and $N/4 = 30$. Which is the Q_1 class?

Class interval	0-10	10-20	20-30	30-40	40-50	50-60
f	8	16	30	36	20	10
c.f.	8	24	54	90	110	120

Options

- A. 20-30
- B. 0-10
- C. 10-20
- D. 30-40

Correct answer: 20-30

Explanation and steps

$N/4 = 120/4 = 30$. The cumulative frequency before 20-30 is 24 and the cumulative frequency up to 20-30 is 54. Since $24 < 30 \leq 54$, the Q_1 class is 20-30.

Question 49 - Practice P1-7

For the following grouped distribution, calculate Q_1 using the grouped-data formula.

For the Q_1 class: $L = 20$, $h = 10$, $f = 40$, $c.f. = 20$ and $N = 160$.

Class interval	10-20	20-30	30-40	40-50	50-60
f	20	40	50	30	20
c.f.	20	60	110	140	160

Correct answer: 25

Explanation and steps

$N/4 = 160/4 = 40$, which lies in the 20-30 class. Using $Q_1 = L + (h/f)(N/4 - c.f.)$, $Q_1 = 20 + (10/40)(40 - 20) = 20 + 5 = 25$.

Question 50 - Practice P1-7

For the following grouped distribution, identify the Q_3 class and calculate Q_3 .

Class interval	0-20	20-40	40-60	60-80	80-100
f	24	48	72	72	24
c.f.	24	72	144	216	240

Options

- A. Q_3 class: 60-80; $Q_3 = 70$
- B. Q_3 class: 40-60; $Q_3 = 50$
- C. Q_3 class: 60-80; $Q_3 = 75$
- D. Q_3 class: 80-100; $Q_3 = 90$

Correct answer: Q_3 class: 60-80; $Q_3 = 70$

Explanation and steps

$3N/4 = 3 \times 240/4 = 180$. Since $144 < 180 \leq 216$, the Q_3 class is 60-80. Using $L = 60$, $h = 20$, $f = 72$ and $c.f. = 144$: $Q_3 = 60 + (20/72)(180 - 144) = 60 + 10 = 70$.